

RAWDAH GATE



DESCRIPTION

Rawdah Appointment System is a hypothetical software system designed to organize the process of booking appointments to visit Al-Rawdah Al-Sharifah at the Prophet's Mosque. The system allows users to create an account, enter their information, view available appointment times, and select a suitable time for their visit.

Users can also view and cancel their appointments when needed. The system administrator can manage available appointment times, organize bookings, and monitor the reservation process.

The main goal of the system is to make the appointment booking process easier and more organized, reduce booking conflicts, and provide users with a simple and clear experience.

OBJECTIVES

- Reduce overcrowding and congestion in prayer areas.
- Manage the number of worshippers according to designated time slots.
- Prevent emergencies and risks caused by overcrowding.

TARGET AUDIENCE

This project targets several groups involved in the appointment booking and management process at the Prophet's Mosque, aiming to meet users' needs and provide organized and efficient access to services.

Visitors to the prophet's mosque

Visitors who want to organize their visit to the Prophet's Mosque and book appointments.

Employees

Employees responsible for managing and organizing appointments and requests.

Security personnel

Security personnel responsible for organizing visitor entry and ensuring compliance with appointments.

NON FUNCTIONAL REQUIREMENTS

For the non-functional requirements, the project will focus on the following aspects:

Usability

The system focuses on ease of use by providing a clear interface and simple essential tasks, helping users complete the booking process without difficulty

Security

The system provides an appropriate level of security by requiring account credentials during login, helping protect user accounts and control access to services.

FUNCTIONAL REQUIREMENTS

Slot Booking

The system allows visitors to view available time slots, choose a suitable appointment, and confirm their booking.

QR Code Permit Generation

The system automatically generates a unique QR code for each confirmed booking.

Schedule and Capacity Management

The system allows administrators to set available time slots, specify the maximum number of visitors, and organize schedules for men and women.

Appointment Cancellation and Modification

Users can cancel or modify their appointments when needed.

Literature Review: Research and Project Application

Research 1 Summary

Crowd Management in Religious Gatherings –Systems Design Approach

The research addresses overcrowding at Al-Rawdah Al-Sharifah due to its limited space and the large number of visitors.

It proposes using digital systems to improve crowd management through:

- Advance booking and scheduled time slots.
- QR codes for visitor verification.
- Capacity limits for each time slot.

Research 1 Application to Rawdah Gate

Area

How We Benefited

Data Collection

Used Process Flow Analysis to identify user steps.

System Design

Applied QFD to translate user needs into technical requirements.

Booking System

Adopted advance booking, time slots, QR codes, and booking modification and cancellation.

Capacity Management

Defined visitor limits and designed an admin dashboard to manage and monitor bookings.

Research 2 Summary

Medical Appointment Booking Application

The research discusses a medical appointment booking application developed to address difficulties faced by patients and staff when booking, changing, or canceling appointments.

Main Features:

- Allows users to select gender, date, and suitable appointment time.
- Reviews similar systems, including HealthEngine in Australia.
- Uses the Prototype Model for iterative development and user feedback.
- Uses diagrams to illustrate system functions and relationships.

Research 2 Application to Rawdah Gate

Area

Application to Our Project

Appointment Booking

Adopted the concept of selecting suitable dates and time slots.

Development Methodology

Apply the Prototype Model to build, test, and improve the system.

System Design

Use diagrams to explain system functions and component relationships.

User Experience

Simplify booking and allow appointment modifications and cancellations.

Research 3 Summary

Toward an Integrated Intelligent Framework for Crowd Control and Management (IICCM)

The research proposes an intelligent framework using AI and IoT to improve crowd management and prevent overcrowding.

Main Technologies:

- Crowd Vision: Uses cameras and computer vision to count visitors and detect bottlenecks.
- Crowd eHealth & IoT: Monitors environmental conditions and health indicators.
- AI Risk Assessment: Predicts congestion and crowd risks before they occur.

Research 3 Application to Rawdah Gate

Area

Application to Our Project

Capacity Verification

Compare actual visitor counts with scanned QR permits.

Dynamic Scheduling

Adjust bookings when congestion or delays are predicted.

Preventive Measures

Verify health clearance during peak times.

System Availability

Consider cloud-based architecture to support high traffic and system availability.

THE END

THANK YOU
FOR LISTENING